When Abstention Fails to Transfer Across an Ethical Benchmark Shift

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Abstract

An assistant may defer cases on which a learned judgment is uncertain. That strategy is useful only if confidence continues to rank errors when the cases change. We prove a sharp transfer bound for risk conditional on a fixed acceptance rule. When target coverage is observed, an exact interval can be tighter. We then fit a lexical classifier on the public ETHICS commonsense corpus and freeze three abstention rules on a development split. At the strictest raw-confidence threshold, accuracy among retained cases rises on the ordinary test but falls on the hard test. The rule also omits every short test case at that threshold. Reporting coverage and error separately exposes both effects. At equal short- and long-text quotas, three confidence rankings beat a score-free baseline on the ordinary split. None beats it on the hard split.

Summary and scope

Question. When can a frozen abstention rule make accepted ethical judgments less reliable after a shift?

Finding. A sharp total-variation bound applies at fixed acceptance. A sharper interval uses the observed coverage. On ETHICS, the strict raw-confidence rule retains zero short texts, and reweighting retained texts by length and label does not explain the hard split's accuracy drop.

Evidence and scope. The lexical experiment is separate empirical evidence on public dataset labels. The result supports checking retained errors and omissions before automating actions. It gives no estimate of language-model behavior or real-world moral correctness.

1 Abstention changes the population being judged

A tool-use assistant may ask for human review when a classifier cannot decide whether an action is permissible. When the classifier accepts a case, the relevant risk is conditional on acceptance. A lower acceptance rate can improve that risk when uncertainty is informative. It can also concentrate mistakes or leave one type of user request entirely without automatic service.

Selective classification formalizes the coverage and error tradeoff [1]. Empirical work on uncertainty under distribution shift shows the need to test confidence on changed inputs [2]. Here we use a deliberately small classifier to isolate what a frozen abstention threshold does on two public splits of an ethical judgment benchmark [3].

2 A sharp conditional transfer bound

Let P and Q be two laws on the joint input and label. Fix a classifier and acceptance event A before evaluating either law. Let E be the classifier's error event, $p = P(A) > 0$, $q = Q(A) > 0$, and $r_P = P(E \mid A)$, $r_Q = Q(E \mid A)$.

Theorem 1. *If* $\mathrm{TV}(P, Q) \le \varepsilon$, *then*

$$|r_Q - r_P| \le \min\left\{1, \frac{\varepsilon}{\max(p, q)}\right\}.$$

Proof. Write $\Delta_1 = Q(A \cap E) - P(A \cap E)$ and $\Delta_0 = Q(A \cap E^c) - P(A \cap E^c)$. Both have absolute value at most ε. Direct expansion gives

$$q(r_Q - r_P) = (1 - r_P)\Delta_1 - r_P\Delta_0.$$

The absolute value is at most ε, since the two nonnegative coefficients sum to one. Reversing the roles of P, Q gives the same bound with denominator p. Combining both and using the unit range proves the claim. □

For sharpness, use three states for retained error, retained correct, and rejection. Let $P = (0, c, 1-c)$ and $Q = (\varepsilon, c-\varepsilon, 1-c)$ with $0 < \varepsilon \le c \le 1$. Both have coverage c and their conditional risks differ by ε/c. Exact enumeration on denominator-eight laws checked all 1,936 ordered pairs with positive coverage and found 560 nontrivial equalities. The theorem concerns joint input and label laws. Text similarity alone cannot certify ε. We do not estimate a deployment shift radius from the corpus.

3 Using target coverage before labels arrive

Coverage can be measured before anyone adjudicates accepted predictions. Let $a = P(A \cap E)$ and write $d = q - p$. Assume that each of the three cells for accepted error, accepted correct, and rejection can receive mass in the target model. Set

$$L(q) = \max\{0, a + [d]_+ - \varepsilon\},$$
$$U(q) = \min\{q, a + \min(0, d) + \varepsilon\},$$

where $[d]_+ = \max(0, d)$.

Theorem 2. *If* $|q - p| > \varepsilon$, *the coverage is incompatible with the assumed total-variation bound. Otherwise the sharp range of retained risk is* $[L(q)/q, U(q)/q]$. *If target coverage is known only to lie in a positive closed interval, first intersect the interval with* $[p - \varepsilon, p + \varepsilon]$. *If the intersection is nonempty, the coverage nearest* p *gives the sharp endpoints over that entire intersection.*

Proof. The changes in the three cell masses are $x - a$, $d - (x - a)$, and $-d$, where $x = Q(A \cap E)$. They sum to zero. For three such changes the total variation is their largest absolute value, since the sum of the positive changes equals the absolute value of the sum of the negative changes. Bounding all three and imposing $0 \le x \le q$ gives $[L(q), U(q)]$. Every allowed triple of masses attains its projected total variation by retaining the source law conditional on each cell of positive source mass. For a source-null cell, use any law supported there. This gives the sharp range in the stated model.

For $q \le p$, the lower ratio is $\max(0, (a - \varepsilon)/q)$ and is nonincreasing in q. The upper ratio is $\min(1, 1 + (a + \varepsilon - p)/q)$ and is nondecreasing. For $q \ge p$, the lower ratio is $\max(0, 1 + (a - \varepsilon - p)/q)$ and is nondecreasing, and the upper ratio is $\min(1, (a + \varepsilon)/q)$ and is nonincreasing. Thus the coverage nearest p gives the widest risk interval among the feasible coverages. □

For example, suppose source coverage is 0.60 with retained risk 0.20. A shift radius of 0.20 and target coverage of 0.40 give the exact retained-risk interval $[0, 0.30]$. The earlier symmetric inequality gives an upper bound of 8/15. The sharper result uses the direction of the coverage change. The verifier agreed with direct enumeration on 3,168 fixed coverage cases and 14,256 interval cases. A measured coverage outside the feasible interval challenges the radius assumption. Coverage inside the interval still leaves the radius unverified.

4 Frozen comparison on public text

The public ETHICS commonsense files contain short and long scenarios with binary judgment labels. The authors' hard split was produced by adversarial filtering [3]. We use their split as a stress condition. Short and long are dataset strata, not demographic groups.

Normalize and hash every text. Drop conflicting duplicates and keep one copy of each remaining normalized text. Assign training texts to fitting or development by hash modulo five. This gives 11,093 fitting texts and 2,816 development texts. Remove any normalized training digest from each official test split. This removes 1,160 ordinary test texts, leaving 2,725. The hard split has 3,964 remaining texts and no exact normalized training overlap. This filter controls exact normalized overlap only. The same public splits have been used for earlier exploratory baselines, so this is a frozen comparison within an existing benchmark rather than a new external blind evaluation.

Fit one multinomial word classifier with additive smoothing. Its 14,269-word vocabulary is selected on the fitting texts. Let s be its log odds and predict the positive label when $s > 0$. For abstention, compare $|s|$, $|s|$ divided by the number of known tokens, and separate $|s|$ thresholds for short and long texts. The prediction itself is identical across the three rules. Calibrate cutoffs on development data to retain 80, 50, or 20 percent, keeping ties. Apply every threshold unchanged to both test splits. The release of these thresholds is a coverage calibration, with no guarantee that it controls error on a changed law.

Rule and development target	Ordinary		Hard	
	Coverage	Retained accuracy	Coverage	Retained accuracy
Forced prediction	100.00	58.79	100.00	45.59
Raw confidence, 50%	26.28	72.21	53.23	40.76
Per-token confidence, 50%	62.83	59.99	47.05	45.63
Group thresholds, 50%	48.99	63.00	47.33	43.28
Raw confidence, 20%	8.40	89.52	18.84	37.08
Per-token confidence, 20%	30.35	57.19	18.26	50.55
Group thresholds, 20%	19.38	64.39	17.94	46.13

Entries are percentages. The receipt includes the 80 percent target.

At the strictest raw threshold the ordinary split retained 229 texts, of which 205 were correct. The hard split retained 747 texts, of which 277 were correct. Both retained zero short texts. Group-specific calibration retained both strata at this target, but still achieved only 328 correct predictions among 711 retained hard texts. Normalizing the score changed the coverage mix and gave 366 of 724 correct at its strictest hard threshold. These rules trade off different coverage and accuracy profiles, and the experiment does not select a winning rule using the test results.

Consider errors in the direction of declaring a text permissible when the dataset marks it as wrong. Among 429 hard texts declared permissible by the strict raw threshold, the dataset labels 275 as wrong. Under forced prediction the corresponding counts are 1,265 of 2,246. The stricter rule makes fewer such errors in absolute terms because it acts less often. The fraction among its

permissive predictions is higher. Both quantities matter when a deployment uses the output to decide whether to act.

5 Does the mix of retained texts explain the drop?

Fix a rule and form cells by crossing the dataset's short and long categories with its binary label. In split $D \in \{O, H\}$, let w_{Dg} be the share of retained texts in cell g and a_{Dg} their accuracy. On common support, the difference in retained accuracy is

$$\sum_g (w_{Hg} - w_{Og}) \frac{a_{Hg} + a_{Og}}{2} + \sum_g \frac{w_{Hg} + w_{Og}}{2} (a_{Hg} - a_{Og}).$$

The two terms separate changes in shares and changes in accuracy within cells. This is the symmetric Kitagawa decomposition [4]. Expanding each summand cancels both cross terms and leaves $w_{Hg}a_{Hg} - w_{Og}a_{Og}$.

A second comparison holds ordinary cell accuracies fixed and gives them the hard split's shares, yielding $\sum_g w_{Hg}a_{Og}$. If a hard cell has no ordinary observations, its term is bounded between zero and its hard share. Every reweighting on ordinary support must lie between the smallest and largest ordinary cell accuracies, because the weights are nonnegative and sum to one.

Retained cell at the strict raw threshold	Ordinary correct/total	Hard correct/total
Long, label 0	118/132	154/349
Long, label 1	87/97	123/398

Reweighting the ordinary accuracies to match the hard split yields 89.55%, compared with the observed 37.08% on the hard split. Even assigning all weight to the lower-accuracy ordinary cell gives 118/132, or 89.39%. The symmetric decomposition assigns -0.71 percentage points to changed shares and -51.73 points to within-cell changes. The strict raw threshold is the only one of the ten evaluated regimes whose hard accuracy falls below every ordinary length-by-label cell. Across the ten regimes, applying the hard shares to the ordinary accuracies gives a higher estimate than the observed hard accuracy. The comparison is descriptive: other features of the texts can change within a length-by-label cell, and the calculation does not identify which feature drove the errors.

6 Comparing rankings at the same coverage

Different cutoffs accept different numbers of short and long texts. To test the ranking itself, we kept the same fitted classifier and gave each rule the same acceptance quota within each length group. A score-free rule orders normalized text digests and serves as a baseline. We compared it with raw confidence, confidence per known token, and a score learned from development-set correctness.

Split the held development fifth into 1,439 cases for learning the ranking and 1,377 for calibrating the cutoffs. Within each length and predicted-label group, divide per-token confidence into five bins using the ranking cases. Rank by the smoothed correctness rate of the corresponding bin. The rate is the number of correct predictions plus one, divided by the bin count plus two. Break all ties by the same text digest order. These rules were fixed before the new scoring run on the previously explored public splits.

In the matched comparison, accept the top fifth of each length group in each test batch, rounding up. This procedure uses the unlabeled batch to set its quota. Flipping every test label left every

selected set unchanged. Each rule retained 422 short and 124 long texts on the ordinary split and 341 short and 452 long texts on the hard split.

Ranking	Ordinary correct of 546	Hard correct of 793
Score-free digest	320	348
Raw confidence	351	346
Per-token confidence	357	342
Development correctness	341	345

The per-token ranking gained 37 correct decisions over the score-free rule on the ordinary split and lost six on the hard split. Learning correctness on development data did not reverse that shift. In the separate threshold-transfer comparison, the cutoffs were fitted only on the calibration half and applied unchanged. Every rule retained both length groups. Across the four rules, hard-split acceptance counts ranged from 320 to 360 for short texts and from 379 to 487 for long texts. A batch quota controls representation directly, while a fixed threshold offers a streaming rule whose coverage can drift. Neither comparison uses a new external holdout.

7 Application and limits

A service can use a confidence score to prioritize human review, provided it checks both coverage and errors under the relevant conditions. Keep authorization and high-impact constraints outside the score. To test a proposed deferral rule, preregister its threshold and evaluate both retained accuracy and omissions in each affected group. A better follow-up would use independent adjudication of actual agent actions on a new cohort and would measure what happens to deferred cases. This study measures agreement with dataset judgments for one small classifier. It supplies no estimate of moral correctness, language-model behavior, or the consequences of real tool use.

The reweighting and decomposition use exact rational arithmetic.

Review status. Independent manuscript review is pending.

References

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